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ORIGINAL RESEARCH

A Study on Migraine in Saudi Arabians Using Pilot Mitochondrial Genome-Wide Association Analysis

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Abstract :

Mitochondrial DNA (mtDNA) mutations have been reported in several neurological diseases and contribute to explaining the pathophysiology of these diseases. Similarly, mutations in her mtDNA may occur in migraine and may explain the effect of reduced ATP production in neurons on triggering migraine attacks. Therefore, the present study aims to investigate the association between mtDNA mutations and migraine in a Saudi population. Subjects and Methods: More than 1950 young Saudi female students were screened for migraine, of which a total of 103 met ICHD-3 criteria. However, 20 cases of migraine were identified at the neurology clinic and agreed to participate in the study. In addition, age-matched 20 healthy controls were also recruited. Mitochondrial sequence variations were excluded from exome sequencing using NCBI GenBank reference sequence NC_012920.1 and analyzed using MITOMAP. Genes with significant single nucleotide polymorphisms (SNPs) were examined for the most significant associated genes using functional enrichment analysis of protein-protein interaction networks with the gene functional classification tools DAVID and STRING 11.5. Genome-wide analysis of mitochondrial sequence variation between migraineurs and controls revealed an association of 30 SNPs in the mitochondrial genome ($p<0.05$). The highest significance ($p = 0.001033$) was observed for coding SNP (rs1603225278) in the CYTB gene and rs386829281 in the origin of replication region. 24 significant SNPs were located in coding regions of 9 genes (ND5, ND4, COX2, COX1, ND3, CYTB, COX3, ND2, and ND1). This is the first study to demonstrate an association between mtDNA mutations and migraine in the Saudi population. His current results serve to highlight the importance of mtDNA mutations in migraine pathophysiology and serve as reference data for large-scale national and international studies.

Keywords : mtDNA variations, CYTB gene, Mitochondrial DNA

Introduction:

Mammalian mitochondria were found to have a specific deoxyribonucleic acid (DNA) in their matrix. Mitochondrial DNA (mtDNA) is responsible for the transcription of proteins essential for mitochondrial function. Paternal her mtDNA is degraded during fertilization, so mtDNA is inherited exclusively from the mother. mtDNA has double strands, but they have different structures. Interestingly, it has been found that mutations can occur in her mtDNA as well as in her nuclear DNA, which can cause the disease.

Additionally, mtDNA is susceptible to mutations due to the lack of histones and high exposure to free oxygen radicals within mitochondria. Mutations in mtDNA can lead to defects in the oxidative phosphorylation process and affect energy production processes.

The first described mtDNA mutation was discovered in 1988 in a patient with mitochondrial myopathy. Several mutations were later discovered and associated with specific diseases [1].

Most mtDNA mutations manifest as neurological symptoms such as myopathy, optic neuropathy, ophthalmoplegia, and myoclonic epilepsy, which occur in diseases such as Leber's hereditary optic neuropathy (LHON) and Kearns-Sayre syndrome (KSS). Subsequently, the study and identification of mtDNA mutations has helped identify the pathophysiology of Alzheimer's disease, Parkinson's disease, diabetes, and cancer. Various forms of migraine were considered to be one of the neurological diseases that could be explained by mtDNA mutations. This is supported by the possible role of increased neuronal excitability, impaired ATP synthesis, and cerebrovascular abnormalities associated with mitochondrial dysfunction in the pathophysiology of migraine. Furthermore, the clear maternal inheritance of migraine may also support the involvement of her mtDNA mutations. Studies examining the pathophysiology of migraine with aura have found some evidence that mitochondrial dysfunction is involved in the first phase of a migraine attack. Similarly, migraine patients have been shown to have higher serum lactate levels during and between attacks, which may further reflect a possible impairment of pyruvate metabolism in the Krebs cycle. A Finnish and German study of migraine patients 10 reported an association with mitochondrial DNA variants [2]. The Nord-Trøndelag Health Study (HUNT) is a single population-based cohort study conducted in Nord-Trøndelag County, Norway and found no significant association between the investigated genetic variants and migraine.

Materials and methods:

The protocol for the current study was developed in accordance with the Ethical Considerations for Human Research of the Declaration of Helsinki and was reviewed and approved by the institution (IRB number: IRB-2021-01-250). Review Imam Abdulrahman Bin Faisal University Board. Young female candidates aged 18-30 years (n = 1950) were tested for migraine, of which a total of 103 met ICHD-3 criteria. Twenty migraine cases among women meeting ICHD-3 criteria were identified at the neurology clinic of University Hospital and from migraine patients at the campus of Imam Abdulrahman bin Faisal University of Dammam (IAU). Participants were recruited after consent was obtained. Saudi Arabia. After obtaining informed consent, age-matched healthy volunteers were enrolled as her controls and had no headache complaints [3].

participants were initially interviewed and were asked about the severity of their headaches on a scale of 1 to 10, number of attacks per month, presence of aura, other associated symptoms, and medication use. Medical history was recorded, including and possible precipitating factors. Stress, lack of sleep, fasting or skipping meals, physical activity, loud noises, changes in weather or temperature, strong smells or lights, certain foods, stages of the menstrual cycle, family history of migraines, etc. Medical history (other chronic illnesses) [4]. The aims and objectives of the study were also explained to them, and they signed a consent form to participate in the study.

Analysis of Mitochondrial Sequence Variations DNA was extracted from blood (QIAamp DNA Blood Mini Kit, Qiagen, Germany), DNA purity was assessed using a Nanodrop, and DNA concentration was tested using a Qubit fluorometer. DNA integrity was checked using agarose gel electrophoresis.

All samples were then sequenced using paired-end whole exome sequencing, followed by quality screening. Quality was considered good if the average sample depth was 7.5 or higher, the sample variant call rate was 0.5 or higher, and the sample average genotype quality was 28 or higher. To filter high-quality variants, criteria such as raw read depth ≥ 10, Phred score quality ≥ 30, and mapping quality ≥ 30 were considered. Python packages were used for quality assessment of samples, genotypes, and variants. Similarly, the Hail standard, based on the Python package, was implemented throughout the genome-wide association pipeline analysis [5].

Results:

To identify mitochondrial sequence variations, age-matched migraine patients (22.10 ± 3.63) and controls (21.86 ± 1.75) were selected for whole exome sequencing. Controls and cases did not differ significantly on 4,444 baseline characteristics, including age (p-value = 0.818), weight (kg) (p-value = 0.115), and BMI (p-value = 0.095). The frequency of the most common precipitating factors in migraine patients is shown in figure 1. Of the migraine patients who participated in this study, 57.9%, 50%, and 55% suffered from aura, used pain medications, and had a family history of migraine (figure 1) [6].

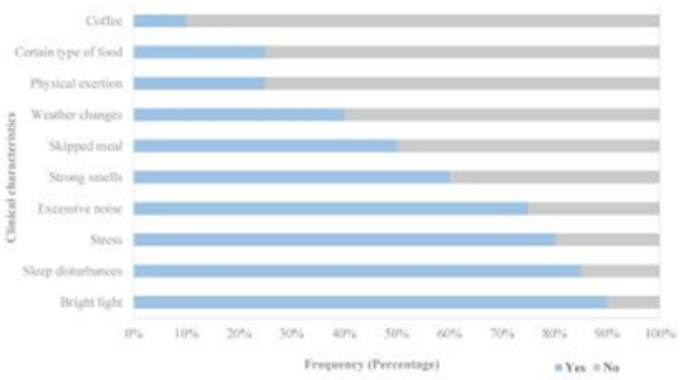

Figure 1: Precipitating possible factors of migraine attack and frequency in the migraine participants from Saudi Arabia.

A total of 189 mitochondrial sequence variations were obtained from exome sequence data. Genome-wide analysis of mitochondrial sequence variation between migraine patients and controls revealed associations of 30 SNPs. ($p<0.05$) in the mictrochondrial genome [7].

The highest significant ($p = 0.001033$) observed in a coding SNP (re 1603225278) in the CYTB gene. Twenty-four significant SNPs are in the coding region of nine different genes (ND5, ND 4, COX2, COX1, ND3, CYTB, COX1, ND2 and ND1) figure 2.

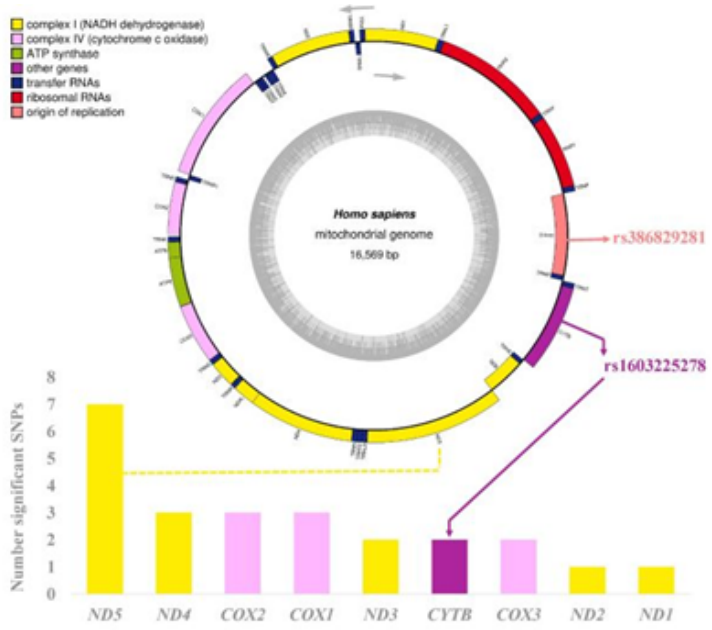

Figure 2: Genes in the mitochondrial and number of significant sequence variations.

Networks of protein-protein interaction analysis and the functional enrichment of the genes (ND5, ND4, COX2, COX1, ND3, CYTB, COX3, ND2 and ND1) with the significant SNPs revealed the most significant associated pathway, oxidative phosphorylation ($p = 2.55\times-12$) in the Kyoto encyclopedia of genes and genomes pathway analysis of the figure 3.

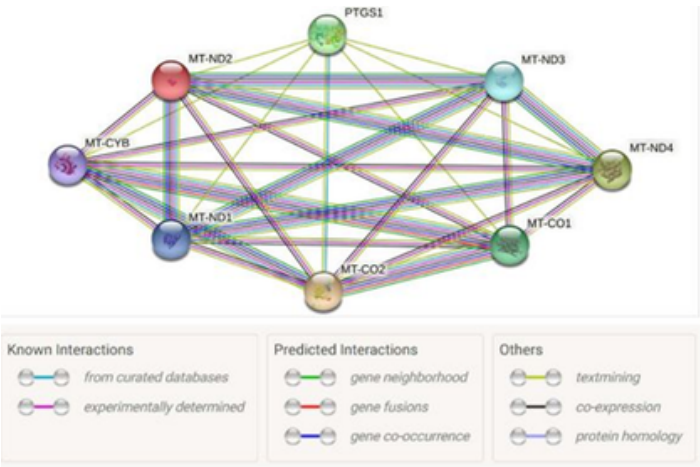

Figure 3: Functional enrichment analysis through protein-protein interaction networks.

This study aimed to identify significant associations between mitochondrial gene mutations and migraine in Saudi candidates using next-generation-based whole mitochondrial DNA sequence analysis. We succeeded in detecting significant associations of 30 different SNPs mainly distributed in nine genes (ND5, ND4, COX2, COX1, ND3, CYTB, COX3, ND2, ND1). The most significant association occurred in her CYTB gene ($p = 0.001033$), while the highest number of associated SNPs was detected in her ND5 gene [8].

These mitochondrial genes are involved in encoding important proteins in the mitochondrial respiratory electron transport chain. The most strongly associated gene, Mt-CYTB, is involved in encoding cytochrome b, which is part of complex III. The Mt-ND genes, including ND1, ND2, ND3, ND4 and ND5, are genes encoding NADH dehydrogenase, and the genes Mt-COX1, mt-COX2 and mt-COX3 encodes cytochrome C oxidase, a subunit of complex IV. Mutations in these mitochondrial genes have been previously reported in several neurological and muscular diseases. MELAS (mitochondrial encephalomyopathy, lactic acidosis, stroke-like episodes), mitochondrial encephalomyopathy, mitochondrial myopathy, Leber's hereditary optic neuropathy, muscle weakness etc. The association of neurological and muscular symptoms with mitochondrial mutations can be determined by their absolute dependence. Organization can be explained. Nerve and muscle tissues regarding aerobic metabolism and mitochondrial oxidative phosphorylation systems. Therefore, disruption of energy production mechanisms, especially the respiratory electron transport chain, can manifest as neurological and muscular symptoms [9].

There are few studies in the literature that focus on investigating the association between mtDNA mutations and migraine. Study identified a mitochondrial transfer RNA mutation (m.3243A>G) in a group of subjects [10]. Further analysis of her individuals from the same cohort showed that subjects with this mutation (m.3243A>G) had a higher prevalence of migraine. Similarly, his study of diseases with mitochondrial mutations showed a higher incidence of migraine in these patients. Research as in other neurological diseases, mitochondrial mutations may contribute to understanding the pathophysiology of migraine [11].

Some studies suggest that changes in energy production due to aberrant oxidative phosphorylation and impairment of the respiratory electron transport chain may reduce ATP levels in involved neurons [12]. In migraine, especially migraine with aura, decreased ATP levels in affected neurons may lower the threshold for cortical spreading depression (CSD), ultimately leading to the induction of aura. Further supporting evidence is that imaging studies have demonstrated mitochondrial impairment in migraine patients. Mitochondrial DNA variations in Arab ancestry have been described to be associated with obesity [13]. The HUNT study using a microarray of known variants conducted in Nord-Trøndelag County, Norway, found no significant association between these specific variants and migraine in the Norwegian population. Additionally, this study also excluded samples that did not pass nuclear genotype quality control, as well as samples with low call rates or close maternal relatives [14,15].

Discussion:

This study is therefore the first to demonstrate an association between mtDNA mutations and migraine in people of Arab ancestry, especially people of Arab descent, are studying the Saudi population, which will add valuable knowledge to the Saudi genome project. Additionally, larger studies are recommended to confirm currently discovered mutations. Further studies on Arab ancestry with larger sample sizes and haplotyping analysis may confirm the potential for a significant impact association on the pathway of migraine onset. Although the small sample size of this study is one notable limitation, the current pilot study opens the possibility of a larger confirmatory study.

Conclusion:

Our current study shows that 30 SNP variants in 9 genes, CYTB, COX1, COX2, ND1, ND2, ND3, ND4, ND5 and COX3, are significantly associated with migraine patients in Saudi Arabia. This should be further confirmed. The most significant SNP variant was located in the CYTB gene, and the highest number of migraine-associated SNP variants was found in ND5. The variant encodes an essential protein that represents a key subunit of the respiratory electron transport chain and therefore plays an important role in energy production and ATP synthesis by neuronal mitochondria. The current study results are the first in this population, and larger studies are needed to confirm the association.

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Conflict of interest:

The authors declare no conflicts of interests in this work.

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